mojaloop

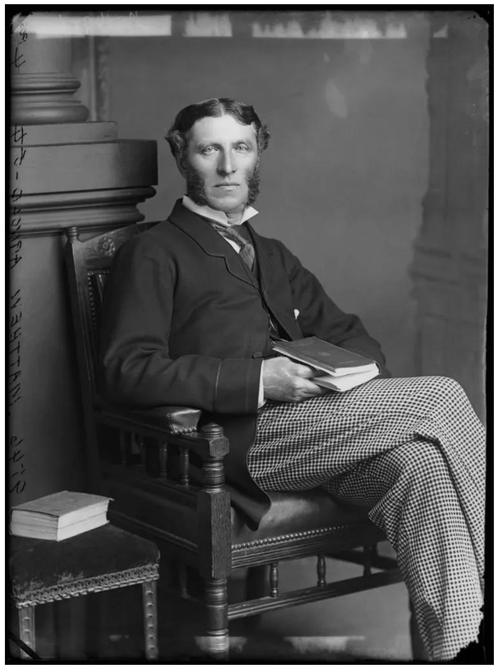
Mojaloop invariants

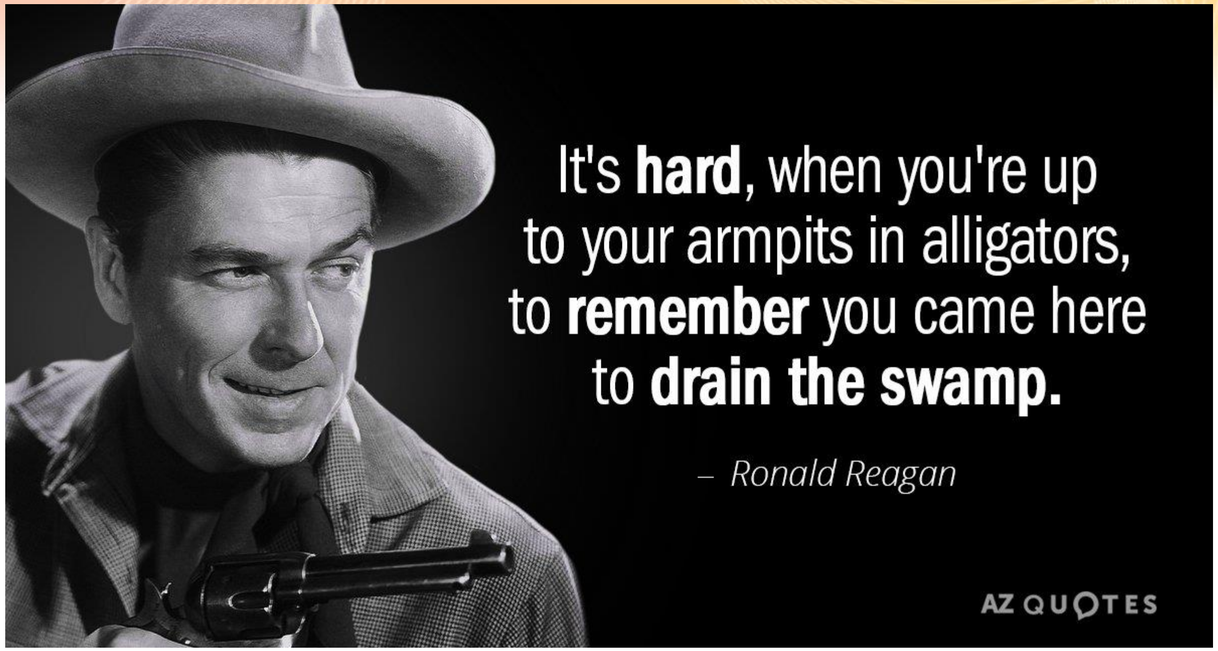
mojaloop

Why invariants?

"I will lift up mine eyes unto the invariants, whence cometh my salvation"

The invariants are **touchstones**…





What do we mean by standards?

Commands which are impossible to reach?



X

Collaborations which are inspirations to excellence?



✓

What are we trying to build?

- A system that connects organisations.
- Those organisations are:
  - Very different from each other
  - Identical in their behaviour
- Our objective: to let them pay other people's customers as easily as they can pay their own:
  - Reliably
  - Immediately
  - Irrevocably



Are all organisations created equal?

Moja, *'tis thine alone, with awful sway,*

To rule Mankind; and make the World obey;

Disposing Peace, and War, thy own Majestick Way.

To tame the Proud, the fetter'd Slave to free;

These are Imperial Arts, and worthy thee.

- Mojaloop wants to help the smallest, simplest account-holding institution process payments as efficiently, reliably and cheaply as an international bank.
- If we can manage that, the poorest customers will get just as good a service as the richest.



Our focus, then:

- The smallest regulated institutions
  - Security
  - Reliability
  - Efficiency
  - Low cost
- How do we do this?
  - Simplification
    - Mojaloop provides the building blocks: schemes use them to build the functions they need.
  - Automation
    - Every time a real person looks at a payment, it costs a poor person money.
  - Cost reduction by design
    - Smart thinking vs. high concept

Simplification

- All payments are the same.
- All payments are push payments.
- The switch never changes the content of messages
- Mojaloop uses a published accounting model to clarify how it manages payments.

Automation

- Agreement of terms before a payment is executed squeezes failure out of the payments process.
  - Enforcing execution according to the terms agreed provides further confidence.
- Timing payments out eliminates fire-and-forget delays and systems clogged with unnecessary queries.
- Using fixed-point arithmetic exclusively means that reconciliation problems are avoided.

Cost reduction: in operation

- Increase participant confidence
  - By guaranteeing settlement of funds.
- Reduce liquidity requirements
  - By settling as often as possible.
- Don’t rely on expensive security techniques
  - Use the public internet and standard encryption techniques.
  - Support participant confidence with non-repudiation signatures.
- Provide guaranteed, secure, tamper-evident central storage for messages
- Don’t rely on long-running conversations.
  - Multiple short conversations are more reliable than fewer long-lasting ones.
  - Idempotency checks stop the system processing retries as if they were new requests.

Cost reduction: the code

- Mojaloop is free, open-source software.
  - 100%, soup to nuts.
- Mojaloop uses standard industry best practice in standard ways.
  - Invention is for implementers.
- Mojaloop is made up of small, easily understood modules
  - Modules have a single function.
- Inter-module communication is
  - Restricted to near neighbours
  - Guaranteed by secure access controls

Performance

- What sort of performance are we looking for?
- Maximum performance come what may?

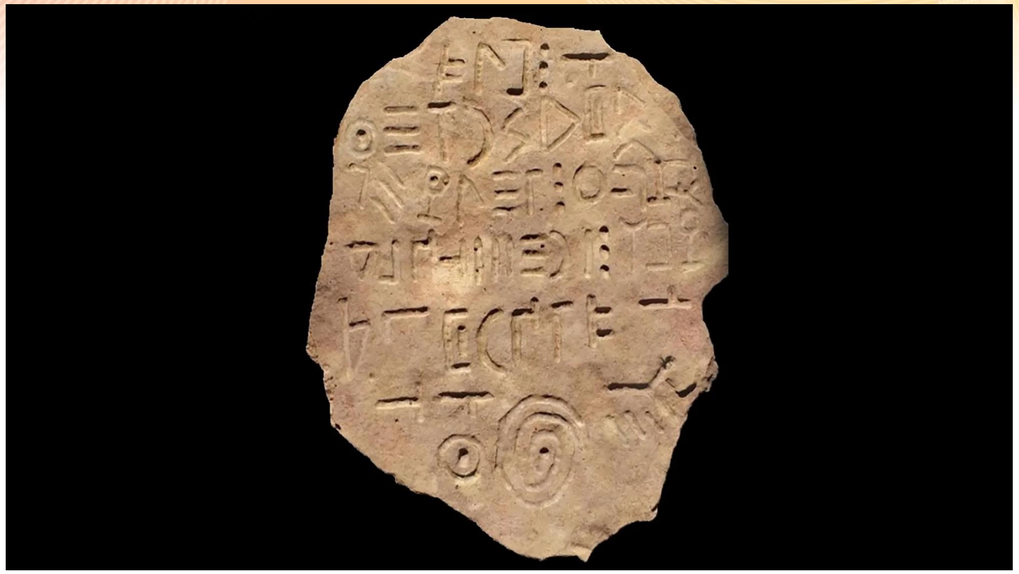




- The best energy/performance ratio?

The future of the invariants

- The invariants are ways of reaching our objectives: they are not the objectives themselves.
- If there's a better way of meeting the objectives, we can change the invariants.
- But they're *our* way of meeting the objectives.
- They carry within themselves the history of the commitment and belief of all those who made Mojaloop what it is today.
- Come and join us on our journey to the future of payments…



An [invariant] is but a paltry thing,
A tattered coat upon a stick, unless
Soul clap its hands and sing, and louder sing
For every tatter in its mortal dress…

W.B.Yeats: *Sailing to Byzantium*

Let's take a closer look at the invariants...

Questions…